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Greater Gastrocnemius Muscle Hypertrophy After Partial Range of Motion Training Performed at Long Muscle Lengths

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Abstract

Kassiano, W, Costa, B, Kunevaliki, G, Soares, D, Zacarias, G, Manske, I, Takaki, Y, Ruggiero, MF, Stavinski, N, Francsuel, J, Tricoli, I, Carneiro, MAS, and Cyrino, ES. Greater gastrocnemius muscle hypertrophy after partial range of motion training performed at long muscle lengths. *J Strength Cond Res* 37(9): 1746–1753, 2023—Whether there is an optimal range of motion (ROM) to induce muscle hypertrophy remains elusive, especially for gastrocnemius. This study aimed to compare the changes in gastrocnemius muscle thickness between calf raise exercise performed with full ROM ($FULL_{ROM}$), partial ROM performed in the initial ($INITIAL_{ROM}$), and final ($FINAL_{ROM}$) portions of the ROM. Forty-two young women performed a calf training program for 8 weeks, 3 days·week^{-1}, with differences in the calf raise ROM configuration. The calf raise exercise was performed in a pin-loaded, horizontal, leg-press machine, in 3 sets of 15–20 repetition maximum. The subjects were randomly assigned to 1 of the 3 groups: $FULL_{ROM}$ (ankle: −25° to +25°), $INITIAL_{ROM}$ (ankle: −25° to 0°), and $FINAL_{ROM}$ (ankle: 0° to +25°), where 0° was defined as an angle of 90° of the foot with the tibia. The muscle thickness measurements of medial and lateral gastrocnemius were taken by means of B-mode ultrasound. $INITIAL_{ROM}$ elicited greater medial gastrocnemius increases than $FULL_{ROM}$ and $FINAL_{ROM}$ ($INITIAL_{ROM}$ = +15.2% vs. $FULL_{ROM}$ = +6.7% and $FINAL_{ROM}$ = +3.4%; $p \leq 0.009$). Furthermore, $INITIAL_{ROM}$ elicited greater lateral gastrocnemius increases than $FINAL_{ROM}$ ($INITIAL_{ROM}$ = +14.9% vs. $FINAL_{ROM}$ = +6.2%; $p < 0.024$) but did not significantly differ from $FULL_{ROM}$ ($FULL_{ROM}$ = +7.3%; $p = 0.060$). The current results suggest that calf training performed at longer muscle lengths may optimize gastrocnemius muscle hypertrophy in young women. Therefore, when prescribing hypertrophy-oriented training, the inclusion of the calf raise exercise performed with partial ROM in the initial portion of the excursion should be considered.

Key Words: resistance training, muscle mass, joint excursion, triceps surae, length-tension relationship

Introduction

Range of motion (ROM) can be operationally defined as the degree of movement at a specific joint during the execution of an exercise (11). Resistance exercise ROM has the potential to modulate muscular adaptations because depending on the specific ROM used in each repetition, factors such as internal moment arm length, the portion of the force-length relationship to which the stimulus is applied, and muscle activation will be differentially affected (6,18). ROM has been classified as full ($FULL_{ROM}$) when a given movement is performed without restriction in the degrees of excursion (13). Notably, ROM may be intentionally limited in the initial ($INITIAL_{ROM}$) or final ($FINAL_{ROM}$) portions of a full ROM, taking the concentric action as reference (13). Despite the widely held belief that $FULL_{ROM}$ is superior to partial ROM to induce muscle hypertrophy; currently, there is not a definitive consensus on this statement (12), with findings suggesting the superiority of full ROM over partial ROM (2), others suggesting similarity (30), and some others suggesting superiority of partial over full ROM (7,29).

When trying to reconcile the findings, it is possible to notice that training at longer muscle lengths, through $FULL_{ROM}$ or $INITIAL_{ROM}$, frequently elicits greater muscle hypertrophy (13). For instance, knee extension exercise performed with $FULL_{ROM}$ and $INITIAL_{ROM}$, taking the concentric muscle action as reference (i.e., at longer muscle length), elicited more favorable muscle hypertrophy in rectus femoris and vastus lateralis than $FINAL_{ROM}$ (i.e., at shorter muscle length) (29). Although not unanimous (7), the superiority of training at longer muscle lengths has been observed in different studies (20,23,24). Among the mechanisms proposed to explain these results is the mechanical muscle characteristics, especially the length-tension relationship (9). Thus, if the fibers of given muscle work, for example, on the descending limb of the length-tension relationship, it may experience additional mechanical tension from passive elements (9). Indeed, this may be the case for the rectus femoris and vastus lateralis, which work on the plateau and descending limb of the length-tension curve (5,35). If accepted, this hypothesis implies that optimal ROM is muscle dependent. Considering that most research has focused on hypertrophy of the quadriceps (17,23,24,29), findings for other muscles are limited or nonexistent, as is the case for the gastrocnemius (13).

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The gastrocnemius muscles are biarticular superficial calf muscles crossing the knee and ankle joints (15). With knees fully extended and ankle dorsiflexed, the medial and lateral gastrocnemius reach longer fascicle lengths than the plantar flexed position (38–46 mm vs. 26–30 mm) (15). Notably, gastrocnemius seems to work on the ascending and plateau portions of the length-tension curve (10,21,39), suggesting that it may produce reduced active force at shorter muscle lengths (33). Also, it implies that the gastrocnemius probably needs to reach longer muscle lengths to produce and experience maximum active force levels (9,16). Theoretically, gastrocnemius would hypertrophy in a greater magnitude when trained in $INITIAL_{ROM}$ than in $FINAL_{ROM}$. As a counterargument, the internal moment arm reduces at longer muscle lengths (21), decreasing the contribution of these muscles to plantar flexor torque. Thus, it remains elusive whether there would be superiority of $INITIAL_{ROM}$ over $FINAL_{ROM}$. Therefore, this study compared the effects of calf training with $FULL_{ROM}$, $INITIAL_{ROM}$, and $FINAL_{ROM}$ on medial and lateral gastrocnemius muscle size changes. We hypothesized that $INITIAL_{ROM}$ could be superior to $FINAL_{ROM}$, but similar to $FULL_{ROM}$.

Methods

Experimental Approach to the Problem

This was a between-group repeated-measures design in which subjects were randomly allocated to 1 of 3 possible ROM configurations in the calf training: $FULL_{ROM}$ (ankle joint: −25° to +25°), $INITIAL_{ROM}$ taking the concentric muscle action as reference (ankle joint: −25° to 0°), or $FINAL_{ROM}$ taking the concentric muscle action as reference (ankle joint: 0° to +25°) training groups, where 0° was defined as an angle of 90° of the foot with the tibia. The current investigation was executed over 12 weeks. Weeks 1, 2, 11, and 12 were used for muscle thickness measurements, whereas the calf training program was performed for 8 weeks (weeks 3–10). All experimental groups trained 3 times a week, totaling 24 training sessions. Each training session was composed of 3 sets of 15–20 repetition maximum in the bilateral calf raise exercise. Subjects were assessed at pretraining and posttraining for medial and lateral gastrocnemius muscle thickness measured with ultrasound imaging.

Subjects

This study sample consisted of apparently healthy adult women aged between 18 and 35 years. Volunteers were recruited through dissemination on social media and folders distributed in points of greater circulation within the campus of the local university. All volunteers underwent a clinical anamnesis and answered the physical activity readiness questionnaire (PAR-Q). The inclusion criteria were not responding "yes" to 1 or more PAR-Q questions; not having osteomyoarticular problems that could compromise the performance of the calf training program; not being a user of dietary supplements or anabolic steroids (self-reported information); and not being engaged in resistance training for at least 6 months before participation in this study. Subjects who did not attend at least 85% of training sessions or missed 2 consecutive training sessions were excluded from the analyses. All subjects were instructed to maintain their routines and eating habits during the intervention. After being informed about the study proposal and procedures, the subjects signed a written informed consent form. Written informed consent was obtained from all subjects after a detailed description of study procedures was provided. This investigation was conducted according to the Declaration of Helsinki, and the project was submitted and approved by the State University of Londrina Research Ethics Committee (3.930.966). Figure 1 illustrates the sampling process, with information on the number of subjects recruited and interviewed, allocation to experimental groups, dropouts throughout the training process, and the final number of subjects who completed the study and were included in the analyses. Forty-two young women [$FULL_{ROM}$, n = 14 (22.0 ± 4.0 years, 69.0 ± 17.6 kg, 165.1 ± 7.7 cm); $INITIAL_{ROM}$, n = 12 (22.5 ± 2.3 years, 61.3 ± 9.2 kg, 162.2 ± 4.5 cm); and $FINAL_{ROM}$, n = 16 (22.3 ± 3.4 years, 64.8 ± 15.0 kg, 161.7 ± 4.8 cm)] ultimately completed the study and were included for final analyses.

Procedures

Gastrocnemius Muscle Thickness Measurement. Medial and lateral gastrocnemius muscle thickness were taken by means of B-mode ultrasound (FIGLABS, FP - 102, SAEVO, SP, Brazil) with a 53-mm, 7.5-MHz linear probe model L741. On arrival at the laboratory, the subject was placed in a prone position on a stretcher and rested for 10 minutes before the beginning of the assessment. A generous quantity of water-soluble transmission gel was applied over the assessed muscle without compressing the skin. Images were obtained on the right leg with the probe perpendicular to the tissue. Image acquisitions of the medial gastrocnemius were taken with the probe positioned in the thickest and more prominent site of the leg from a posteroanterior view (27). Measurements of the lateral gastrocnemius were taken at the proximal third between the lateral epicondyle of the femur and the lateral malleolus of the fibula (27). Two assessors participated in measurement procedures, with the first handled the probe and the second responsible for freezing the images. When the quality of the image was deemed to be satisfactory, the second assessor obtained muscle thickness dimensions using the machine's calculation package. The second assessor was blinded to group allocation. Muscle thickness of medial and lateral gastrocnemius was defined as the distance from the superficial to deep aponeuroses that borders the soleus. Reference lines were drawn on the subject's skin with a dermatographic pen on the sites at which images were obtained and were reinforced and maintained throughout the study to ensure that the measurement was taken at the same place at baseline and posttraining. The muscle thickness assessments were performed in the morning hours (7–11 AM) before and after training. During baseline assessments, 17 subjects were randomly chosen to be evaluated on 2 days separated by 72 hours to determine the reliability of the muscle thickness measurements. The intraclass correlation coefficients for the medial and lateral gastrocnemius were 0.993 and 0.990; the coefficients of variation were 1.4 and 1.6%; the standard errors of measurement were 0.028 and 0.029 cm, respectively. Posttraining measurements were performed at an interval of 72–96 hours after the last training session.

Calf Training Program. The resistance training program was performed 3 times per week (Mondays, Wednesdays, and Fridays) in the afternoon for 8 weeks. The calf raise exercise was performed bilaterally, in a pin-loaded horizontal leg-press machine (Ipiranga, Presidente Prudente, SP, Brazil) in 3 sets of 15–20 repetitions, executed until momentary concentric failure (i.e., when the subject has reached the point where, despite trying to do so, they cannot complete the concentric muscle action of their current repetition). This repetition range was chosen based

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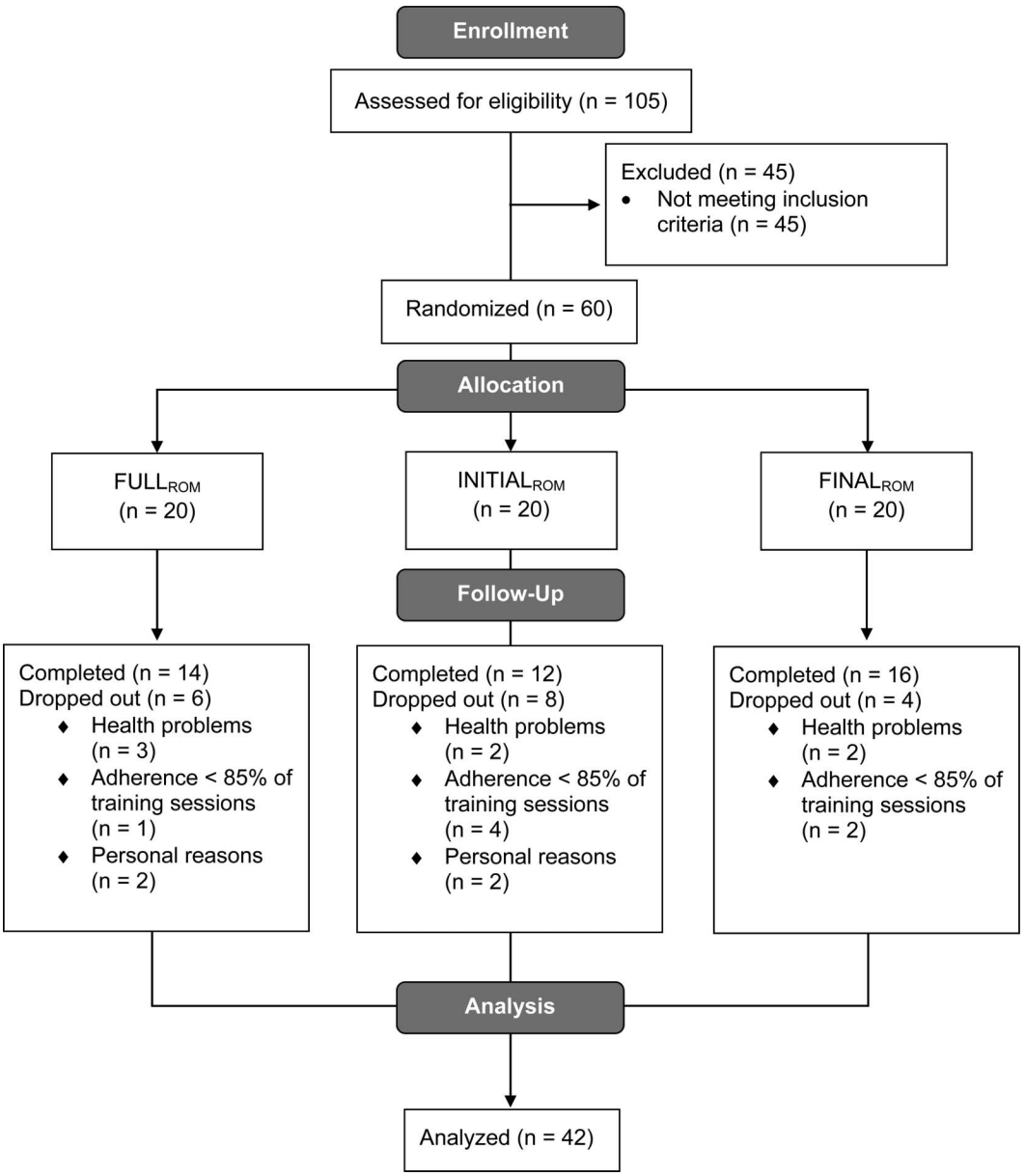

Figure 1. CONSORT flow chart. $FULL_{ROM}$ = full range of motion, $INITIAL_{ROM}$ = initial part of the range of motion, $FINAL_{ROM}$ = final part of the range of motion.

on their common inclusion in calf hypertrophy–oriented training programs (27,31). Whenever the subject reached 20 repetitions in a given set and reported that they could perform ≥1 repetition, the weight was increased by 2–5% for the next set to ensure that the subjects kept performing the sets to (or very near to) failure in the established repetition range. The rest period was 60–90 seconds between the sets. The $FULL_{ROM}$ group performed calf raise exercise with no degree of movement restrictions (ankle joint: −25° to +25°; 0° = foot 90° relative to the tibia). $INITIAL_{ROM}$ performed calf raise in the partial ROM at the initial excursion of the concentric muscle action of plantar flexion (ankle joint: −25° to 0°). $FINAL_{ROM}$ performed calf raise in the partial ROM at the final excursion of the concentric muscle action of plantar flexion (ankle joint: 0° to +25°). An illustration of the training groups can be seen in Figure 2. These ROMs were defined from pilot data collection in which we observed that with overload (i.e., during calf raise exercise), the complete ROM was approximately 50°, and initial and final ROMs were approximately 25°. The ankle joint angles of −25° and +25° were validated through the displacement of the weight stack on the stem. Using an inelastic tape measure and a goniometer, we identified that 6 cm of weight stack displacement corresponds to 25° of ankle flexion. This weight stack displacement was the parameter adopted to validate the specific ROM. The research assistants confirmed when the subject had reached the required weight stack displacement during the calf raise exercise. In the $INITIAL_{ROM}$ group, a metallic structure was placed in the stem where the weight stack of the machine run to serve as a limiter at the top part of desired plantar flexion angle (0°). In the $FINAL_{ROM}$ group, a step in EVA was used to serve as a limiter at the bottom part of desired plantar flexion angle (0°). In $INITIAL_{ROM}$ and $FINAL_{ROM}$ groups, the 0° angle was identified using a goniometer. The supplementary material (http://links.lww.com/JSCR/A383) illustrates how the mechanical stops were used to delimit the ROM excursion. The

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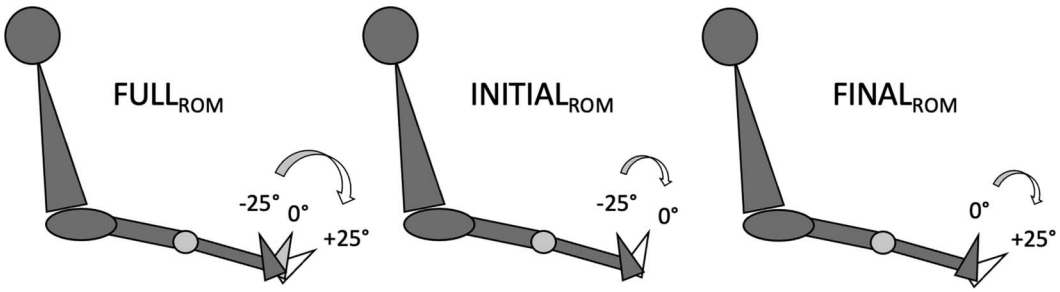

Figure 2. Illustration of 3 experimental groups. FULL$_{ROM}$ = full range of motion, INITIAL$_{ROM}$ = initial part of the range of motion, FINAL$_{ROM}$ = final part of the range of motion.

subjects performed the calf raise exercise at a tempo of 1:2 seconds (concentric and eccentric phases, respectively), with the knee extended and the foot positioned on the platform supported by metatarsals. Subjects performed all sessions under specialized supervision (1:2 subject:supervisor ratio) to ensure safe and proper execution. All weights and repetitions performed were recorded. The volume load was calculated as the number of sets × number of repetitions × load lifted. The average volume load of the first training sessions in week 1 was used as each subject's reference value. Thereafter, volume load progression was calculated based on percentage differences between the average volume load produced between training sessions 1 and 2–24.

Statistical Analysis

Data distribution and variance homogeneity were verified through Shapiro-Wilk and Levene's tests, respectively. The one-way analysis of variance (ANOVA) was used to compare the baseline characteristics between the groups. The comparison of different ROM configurations (FULL$_{ROM}$ vs. INITIAL$_{ROM}$ vs. FINAL$_{ROM}$) effects on gastrocnemius muscle thickness was made using an analysis of covariance (ANCOVA) of the raw difference between baseline and posttraining measures with the baseline score as a covariate. When the *F*-ratio was significant, Bonferroni's post hoc test was used to identify the differences between pretraining and posttraining raw data. The *p* values for group comparisons were also presented. The interpretation of the effect of time was made from the 95% confidence interval (95% CI) of the mean difference from pretraining to posttraining (i.e., there was a significant difference when the inferior and superior confidence limits did not cross zero). The effect size (ES) was calculated as posttraining group mean minus the pretraining mean, divided by the groups-pooled pretraining standard deviation (4). The one-way ANOVA was used to compare the initial volume load between the groups. The volume-load progression slopes were generated using linear regression and compared through an *F*-test. For all statistical analyses, significance was accepted at $p < 0.05$. The data were stored and analyzed using JASP software (version 0.14.1, Amsterdam, NL). The data are presented as mean and standard deviations.

Results

No significant between-group differences for subject characteristics or dependent variables were detected at baseline ($p \geq 0.265$). Session attendance was 93.4 ± 5.7% for the FULL$_{ROM}$, 91.0 ± 5.6% for the INITIAL$_{ROM}$, and 92.0 ± 5.8% for the FINAL$_{ROM}$ with no difference between the groups.

Table 1 displays the pretraining and posttraining values of gastrocnemius muscle thickness. There were significant increases in muscle thickness of the medial gastrocnemius for FULL$_{ROM}$ and INITIAL$_{ROM}$ ($p \leq 0.001$) but not for FINAL$_{ROM}$ ($p = 0.053$). A significant group effect was observed for the changes in the medial gastrocnemius muscle thickness ($F = 9.254$; $p < 0.001$). There was a greater increase for INITIAL$_{ROM}$ than for FULL$_{ROM}$ [mean$_{diff}$ = 0.13 cm (95% CI: 0.03, 0.23), $p = 0.009$] and greater increases for INITIAL$_{ROM}$ than for FINAL$_{ROM}$ [mean$_{diff}$ = 0.17 cm (95% CI: 0.07, 0.27), $p < 0.001$], but no difference was observed between FULL$_{ROM}$ and FINAL$_{ROM}$ [mean$_{diff}$ = 0.04 cm (95% CI: −0.05, 0.13), $p = 0.892$]. Figure 3 shows the relative changes on muscle thickness of medial gastrocnemius (FULL$_{ROM}$ = +6.7%; INITIAL$_{ROM}$ = +15.2%; and FINAL$_{ROM}$ = +3.4%).

There were increases in muscle thickness of the lateral gastrocnemius for FULL$_{ROM}$, INITIAL$_{ROM}$, and FINAL$_{ROM}$ ($p \leq 0.005$). A significant group effect was observed for the changes in the lateral gastrocnemius ($F = 4.464$; $p = 0.018$). There was a greater increase for INITIAL$_{ROM}$ than for FINAL$_{ROM}$ [mean$_{diff}$ = 0.13 cm (95% CI: 0.02, 0.25), $p = 0.024$], but not significant difference was observed between INITIAL$_{ROM}$ and FULL$_{ROM}$ [mean$_{diff}$ = 0.12 cm (95% CI: −0.01, 0.24), $p = 0.060$] and between FULL$_{ROM}$ and FINAL$_{ROM}$ [mean$_{diff}$ = 0.01 cm (95% CI: −0.10, 0.12), $p = 0.999$]. Figure 3 shows the relative changes on muscle thickness of lateral gastrocnemius (FULL$_{ROM}$ = +7.3%; INITIAL$_{ROM}$ = +14.9%; and FINAL$_{ROM}$ = +6.2%).

The groups started the training program with similar volume load (FULL$_{ROM}$ = 2,696.4 ± 460.9 kg, INITIAL$_{ROM}$ = 2,687.0 ± 467.8 kg, FINAL$_{ROM}$ = 2,696.4 ± 460.9 kg; $F = 0.78$, $p = 0.469$). Volume load progression slopes differed significantly between FULL$_{ROM}$ and INITIAL$_{ROM}$ ($F = 63.61$; $p < 0.001$) and between FULL$_{ROM}$ and FINAL$_{ROM}$ ($F = 21.86$; $p < 0.001$) but not between INITIAL$_{ROM}$ and FINAL$_{ROM}$ ($F = 2.019$; $p = 0.162$). Figure 4 shows the volume load progression from session to session in the FULL$_{ROM}$, INITIAL$_{ROM}$, and FINAL$_{ROM}$ groups.

Discussion

The purpose of this study was to examine the effects of 3 different ROM configurations (FULL$_{ROM}$, INITIAL$_{ROM}$, and FINAL$_{ROM}$) on the muscle thickness of gastrocnemius muscles in untrained young women. The main finding of this study is that ROM can influence the magnitude of increases in muscle

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Table 1

Medial and lateral gastrocnemius muscle thickness (in centimeters) before and after 8 wk of the calf raise training with different ROM configurations.*

Muscle thickness	$FULL_{ROM}$	$INITIAL_{ROM}$	$FINAL_{ROM}$
Medial gastrocnemius			
Pre	1.65 ± 0.29	1.58 ± 0.26	1.75 ± 0.27
Post	1.76 ± 0.30†	1.82 ± 0.27†,‡,§	1.81 ± 0.29
$Mean_{diff}$	0.10 (0.04, 0.17)	0.23 (0.16, 0.31)	0.06 (−0.01, 013)
ES	0.40	0.88	0.22
Lateral gastrocnemius			
Pre	1.65 ± 0.32	1.61 ± 0.29	1.62 ± 0.27
Post	1.77 ± 0.34†	1.85 ± 0.34†,‡	1.72 ± 0.25†
$Mean_{diff}$	0.12 (0.04, 0.20)	0.24 (0.15, 0.32)	0.10 (0.03, 0.18)
ES	0.41	0.82	0.34

*Note. ROM = range of motion; $FULL_{ROM}$ = full range of motion (n = 14); $INITIAL_{ROM}$ = initial part of the range of motion (−25° to 0°) (n = 12); $FINAL_{ROM}$ = final part of the range of motion (0° to +25°) (n = 16); ES = effect size. Pretraining and posttraining data are presented as mean and standard deviation, whereas $mean_{diff}$ as mean and 95% confidence intervals.
†$p < 0.05$ vs. baseline.
‡$p < 0.05$ vs. $FINAL_{ROM}$.
§$p < 0.05$ vs. $FULL_{ROM}$.

thickness of the gastrocnemius. Specifically, the $INITIAL_{ROM}$ induced greater gains on the medial and lateral heads compared with the $FINAL_{ROM}$. Moreover, $INITIAL_{ROM}$ elicited greater increases in the medial gastrocnemius than $FULL_{ROM}$. This indicates that partial ROM training at longer muscle lengths can optimize hypertrophy of the gastrocnemius muscles. Our initial hypothesis that $INITIAL_{ROM}$ would be superior to $FINAL_{ROM}$ was confirmed. Conversely, the assumption that there would be similar gains between $INITIAL_{ROM}$ and $FULL_{ROM}$ was not confirmed, given that there was more favorable gastrocnemius muscle hypertrophy for $INITIAL_{ROM}$ configuration. Potential mechanisms and explanations for our findings are discussed following.

In this study, we observed more favorable medial and lateral gastrocnemius hypertrophy for the $INITIAL_{ROM}$, especially when contrasted with $FINAL_{ROM}$ (between-group ES: 0.48–0.66). A possible explanation for this superiority may lie in the muscle length at which the gastrocnemius muscles were trained. Growing evidence suggests that resistance exercises performed at long muscle lengths promote superior muscle hypertrophy (19,20,32). In this study, the calf raise exercise was performed with the knees extended and, specifically in $INITIAL_{ROM}$, the ankle moved exclusively in dorsiflexion angles, in which the gastrocnemius heads reach longer muscle lengths (15). Given that gastrocnemius work on the ascending and plateau portions of the length-tension curve (10,21), it is possible to propose that the gastrocnemius fibers may have produced maximum active force levels from contractile elements (i.e., actin and myosin filaments) in the $INITIAL_{ROM}$ configuration to reach longer muscle lengths (9,16). Thus, gastrocnemius muscle fibers probably experienced optimal mechanical tension that resulted in a more favorable muscle hypertrophy stimulus (38). Conversely, the gastrocnemius fibers in the $FINAL_{ROM}$ group may not have produced maximum active force because of the shorter muscle length in this ROM configuration (15,16). Thus, experiencing reduced mechanical tension and less favorable muscle hypertrophy stimulus (38). This conceivably helps to explain the greater muscle hypertrophy in the $INITIAL_{ROM}$ group.

Still based on the length-tension relationship, it is possible to suggest that gastrocnemius fibers in the $INITIAL_{ROM}$ configuration potentially experienced additional passive tension from elastic elements (e.g., titin), and this may have contributed to eliciting superior muscle growth because of the potential additive effects of stretch and contraction (3,13,25). This phenomenon has been called stretch-mediated hypertrophy (24). Indeed, it has been suggested that there is an increase in titin stiffness (especially in the PEVK segment) in the active muscle when reaching longer muscle lengths that result in further passive tension in the sarcomere (25). Thus, this would result in greater overall mechanical tension, a primary hypertrophic stimulus (38). Importantly, it has been argued that only muscles that work in the descending limb of the length-tension curve appear to experience stretch-mediated hypertrophy (13,28). Although there are findings from a cadaver study suggesting that gastrocnemius may work in the descending limb (5), it remains questionable whether, in fact, the sarcomeres of the gastrocnemius muscle fibers reach the descending limb in vivo human muscles (10,21,39). Therefore, it remains to be determined whether gastrocnemius fibers work in descending limb of the length-tension curve and may experience stretch-mediated hypertrophy.

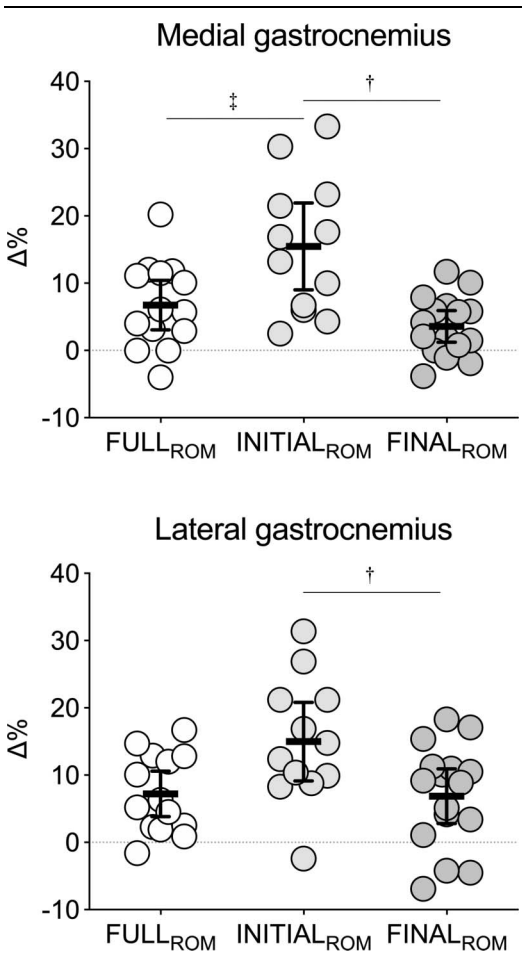

Figure 3. Changes from pretraining to posttraining period for medial and lateral gastrocnemius muscle thickness. †$p < 0.05$ vs. $FINAL_{ROM}$; ‡$p < 0.05$ vs. $FULL_{ROM}$. The horizontal lines represent mean and 95% confidence intervals, whereas each circle represents a subject. $FULL_{ROM}$ = full range of motion; $INITIAL_{ROM}$ = initial part of the range of motion; $FINAL_{ROM}$ = final part of the range of motion.

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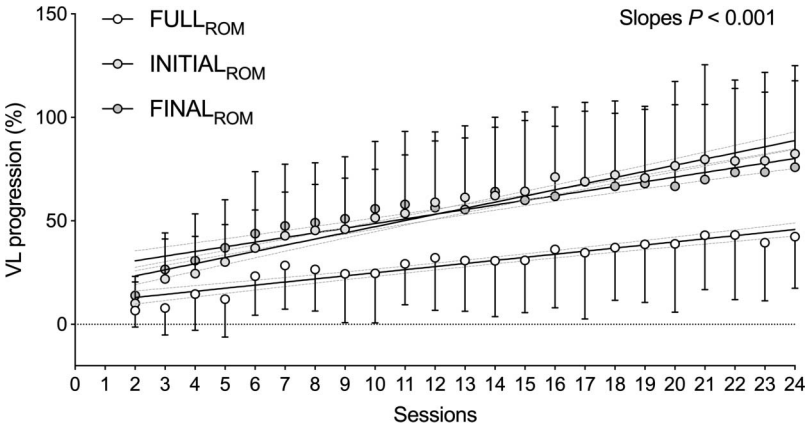

Figure 4. Volume load progression per session for the FULL$_{ROM}$, INITIAL$_{ROM}$, and FINAL$_{ROM}$ configurations, with slopes (continuous straight lines) and 95% confidence intervals (dotted lines). FULL$_{ROM}$ = full range of motion; INITIAL$_{ROM}$ = initial part of the range of motion; FINAL$_{ROM}$ = final part of the range of motion.

Notably, other factors such as internal moment arm and muscle activation play an important role in muscle capacity for producing force and then potentially influence the hypertrophic stimulus (13). Theoretically, the longer the internal moment arm length, the greater the capacity of muscle in producing a turning force at a specific joint (36). Some research has found that internal moment arm lengths increase with increasing plantar flexion angle, thereby being greater at shorter muscle lengths (21,22). This could result in greater force production at a more flexed plantar position and then a more favorable hypertrophic stimulus for the FINAL$_{ROM}$ configuration. However, our results do not support this notion. Thus, it is possible to suggest that the change in the internal moment arm length at different ankle joint angles may not be sufficient to substantially affect the gastrocnemius production of plantar flexion torque. Importantly, investigations have been less devoted to understanding potential differences in the internal moment arm lengths of each muscle and how these might alter with changing ankle joint angles. Thus, further research is needed to determine the internal moment arm lengths of individual muscles and explore the potential influence of this factor on muscle hypertrophy.

Regarding muscle activation at different ankle positions, findings are divergent. For example, there is a report suggesting that surface electromyographic (sEMG) amplitude is subtly greater for lateral gastrocnemius, but not for medial gastrocnemius, at a more dorsiflexed ankle position (i.e., longer muscle lengths) compared with a more plantar flexed position (i.e., shorter muscle lengths) (1). Conversely, other studies found no differences in sEMG amplitude when comparing a more dorsiflexed versus a more plantar flexed ankle position (8,34). Interestingly, studies consistently observe greater plantar flexion torque in more dorsiflexed position (1,8,21). Based on greater gastrocnemius hypertrophy in the INITIAL$_{ROM}$ group observed in this study, and not consistent findings on sEMG, it is possible to suggest that the muscle activation, per se, may not indicate the hypertrophic potential when comparing training at different muscle lengths (37). In fact, torque production across a range of joint angles creates discordant muscle sEMG amplitudes (37). By contrast, greater hypertrophy is commonly observed when training at longer compared with shorter muscle lengths (12,13). Importantly, because of the divergent findings on muscle activation at different ankle positions, further studies are needed to characterize this aspect and investigate the potential relationship between muscle activation and gastrocnemius muscle hypertrophy. In addition, other techniques (e.g., magnetic resonance imaging, ultrasonography) have been used to infer the magnitude of the stimulus in a given muscle and should be explored.

Interestingly, the INITIAL$_{ROM}$ group elicited significantly greater increases in medial gastrocnemius muscle size than FULL$_{ROM}$ (between-group ES: 0.48), and although we did not observe significance for the lateral gastrocnemius, the ES favored the INITIAL$_{ROM}$ (between-group ES: 0.41). Our findings are in accordance with a recent study, which found more favorable muscle hypertrophy of the vastus lateralis and rectus femoris in the INITIAL$_{ROM}$ than in the FULL$_{ROM}$ when performing the knee extension exercise (29). A potential explanation for such INITIAL$_{ROM}$-favorable findings over FULL$_{ROM}$ may be an interaction between muscle length and volume load progression (which influences mechanical tension). In fact, higher rates of volume-load progression seems to result in greater muscle hypertrophy (26). The analysis of the training loads in this study showed that both partial ROM groups (INITIAL$_{ROM}$ and FINAL$_{ROM}$) progressed more than the FULL$_{ROM}$ (see Figure 4). Notably, the INITIAL$_{ROM}$ achieved greater increases in gastrocnemius muscle thickness than the other 2 ROM configurations. Because the magnitude of volume load progressions was similar for the INITIAL$_{ROM}$ and FINAL$_{ROM}$, this seems to indicate that muscle hypertrophy seems to be optimized through an interaction between volume-load progression—thus, overload—and training at longer muscle lengths.

Certainly, our study has limitations that need to be addressed. First, dietary intake and daily physical activity levels were not assessed and remain uncertain whether these factors could exert some influence on muscular adaptations. Importantly, the subjects were instructed to maintain their nutritional habits and not perform any additional exercise on a

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systematic basis. Second, the training intervention in this study lasted 8 weeks; it would be interesting to have longer training periods to verify the effects of different ROM configurations on muscular adaptations. Third, because our study exclusively investigated the medial and lateral gastrocnemius muscle hypertrophy, future investigations should consider examining the influence of ROM on the architectural (e.g., fascicle length and pennation angle) and functional (e.g., maximum voluntary isometric contraction and 1 repetition maximum) parameters, as well as the effects of specific ROM on muscle size changes of the soleus and other muscles with limited or nonexistent findings such as the hamstrings and pectoralis major. Moreover, given that a resistance training program comprises a variety of exercises (14), it is necessary to investigate the effect of different ROM configurations on other exercises that target the calf muscles (e.g., seated calf raise). Finally, this experiment was performed on untrained young women and therefore remains to be determined whether such findings are also observed in other populations of different ages, sex, and training status.

Practical Applications

Although $FULL_{ROM}$ configuration significantly increased both gastrocnemius heads, $INITIAL_{ROM}$ elicited more favorable muscle hypertrophy, especially compared with $FINAL_{ROM}$ configuration. Therefore, from the results of our study, it is possible to suggest that calf raise exercise performed with partial ROM in the initial portion of the movement (i.e., training at longer muscle lengths) may induce greater hypertrophy of the gastrocnemius muscles following 8 weeks of resistance training in untrained young women. These results add to the growing evidence suggesting that resistance training performed at long muscle lengths promotes superior muscle hypertrophy. From a practical perspective, strength and conditioning professionals and practitioners aiming to optimize medial and lateral gastrocnemius muscle growth should consider including this strategy in the hypertrophy-oriented training program.

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